

Big Data & SQL

INSTANT

Interactive Dashboards

Data Analysis Diploma • Session 01

DATA ANALYSIS FUNDAMENTALS

Your Complete Roadmap to Become a Job-Ready Data Analyst

8 Stages

Curriculum Roadmap
From zero to job-ready

7 Modules

Applied Core Modules
Real-world projects

5 Core Tools

Industry Tech Stack
Excel • SQL • Python • BI •
Tableau

Predictive Analytics

Start Session 01 →

Python & Statistics

— Definition & The 5 Core Activities

What is Data Analysis?

The process of turning raw data into decisions that matter.



INDUSTRY DEFINITION

Data Analysis is the process of collecting, cleaning, transforming, analyzing, and interpreting data to discover useful insights and support decision-making.

KEY ACTIVITIES



Collect Data

Gather raw data from multiple sources: files, databases, APIs, and forms.

Stage 01



Clean Data

Fix errors, remove duplicate records, and handle missing null values.

Stage 02



Analyze Data

Apply statistics, logic, and models to find patterns and trends.

Stage 03



Visualize Results

Turn complex numbers into clear, readable charts and interactive dashboards.

Stage 04



Generate Insights

Translate analytical findings into actionable recommendations for the business.

Stage 05

— Business Impact & Value Creation

Why Data Analysis is Important?

Every strong business decision starts with reliable data.



Better Decision Making

Replace guesswork and gut feelings with evidence-based strategic choices.



Understanding Customers

Reveal behavior patterns, purchasing journeys, and product preferences.



Finding Opportunities

Spot untapped market niches and uncover hidden business gaps before competitors.



Improving Performance

Track key performance indicators (KPIs) to optimize operational efficiency.



Predicting Trends

Forecast future demand and market dynamics based on robust historical patterns.

— Everyday Applications Across Industries

Data in Real Life

Data is everywhere — every industry runs on it.



Business

Sales transactions, customer churn, financial records, and supply chain logistics.

● System Standard



Healthcare

Patient electronic health records, treatment efficacy, and hospital capacity planning.

● System Standard



Technology

User clickstream interactions, server logs, API telemetry, and app retention metrics.

● System Standard



Social Media

Posts, comments, sentiment analysis, engagement rates, and viral content reach.

● System Standard

— The Value Hierarchy from Raw Facts to Action

Data vs Information vs Insight

Three stages of the same raw material, each more valuable than the last.



1. Data

Level 1 • Raw Facts

Unprocessed facts, figures, and raw measurements with no context on their own.

EX: "Raw sales numbers: 1400, 2900, 5100"



2. Information

Level 2 • Structured Context

Data organized into a structured, readable report with clear context and labels.

EX: "Monthly sales report: Branch Cairo total revenue is \$150,000"



3. Insight

Level 3 • Actionable Value

The "why" behind the numbers — actionable business meaning that steers decision-making.

EX: "Sales increased 25% due to the targeted social media promotional campaign"

The Two Fundamental Data Categories

Types of Data

Every dataset is built from these two fundamental categories.

DATA CLASSIFICATION SYSTEM:

All Types

Qualitative Data

Quantitative Data



CATEGORICAL • QUALITATIVE

Qualitative Data

Non-Numerical

Descriptive characteristics, labels, and groups that cannot be mathematically calculated directly.

NOMINAL (NO ORDER)

Categories with no intrinsic order

Cities (London, NY)

Colors (Red, Blue)

Gender (M/F)

ORDINAL (ORDERED)

Categories with meaningful rank

Rank (Low, Med, High)

Rating (1 to 5 Stars)

Education Level

Target Visuals: Bar Charts, Donut, Frequency

Categorical



NUMERICAL • QUANTITATIVE

Quantitative Data

Calculable Numbers

Measurable numerical values that support full mathematical calculations and statistical aggregations.

DISCRETE (COUNTABLE)

Countable whole integers

Order Count (42)

Employees (150)

Page Visits (8)

CONTINUOUS (MEASURABLE)

Measurable infinite decimals

Revenue (\$1,249.50)

Duration (4.25 min)

Target Visuals: Histograms, Line, Scatter, KPIs

Numerical

Data Architecture & Schema Organization

Structured vs Unstructured Data

How organized is the data before it reaches an analyst?



TABULAR / SQL

STRUCTURED

Highly organized data stored in standardized schemas, easily queryable with SQL and Excel.

- SQL Database Tables
- Excel Spreadsheets
- Clean CSV Datasets
- Enterprise ERP Systems

Pillar 01

01



JSON / APIS

SEMI-STRUCTURED

Data that does not conform to strict tabular schemas but carries self-describing tags and keys.

- JSON Web API Payloads
- XML Documents
- Web Server Access Logs
- NoSQL Collections (MongoDB)

Pillar 02

02



~80% OF WORLD DATA

UNSTRUCTURED

Data with no predefined conceptual model, representing ~80% of all generated enterprise data.

- Images & Graphic Files
- Video & Audio Streams
- Emails & Customer Chats
- PDF Reports & Documents

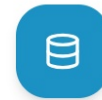
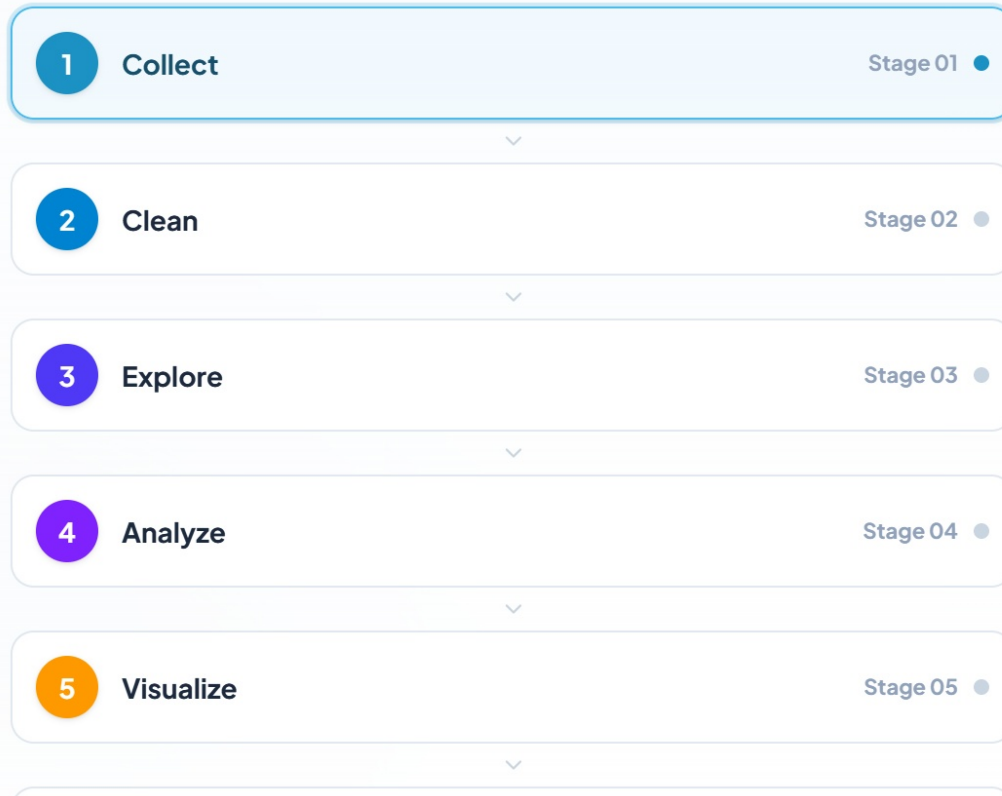
Pillar 03

03

The 6 Sequential Stages of Every Analytics Project

Data Analysis Lifecycle

Every analysis project moves through the same six stages.



STAGE 01 DEEP DIVE

Collect

Identify and extract raw data from internal and external sources: databases, files, APIs, and forms.

KEY OUTPUTS & DELIVERABLES:

✓ Identify Sources

✓ Raw Ingestion

✓ Access Verification

Click any stage on the left to inspect

1 / 6

End-to-End Architecture from Ingestion to Decision

Data Pipeline

How raw data travels from source to decision.

TIER 1: DATA INGESTION & ETL PIPELINE

ETL PROCESS



01

Data Sources

APIs, CSV, Logs



02

Extract

Ingest raw streams



03

Transform

Cleanse & structure



04

Load

Load to warehouse



Flows into Analytics & Consumption

TIER 2: STORAGE, ANALYTICS & DECISION

BUSINESS VALUE



05

Database

SQL Warehouse



06

Analysis

Modeling & queries



07

Dashboard

Power BI & Tableau



08

Decision

Data-driven action

Professional Learning Roadmap

8 Connected Milestones from Zero to Job-Ready

< 01 / 08 >



Math & Stats



Excel



Python Core



Pandas / EDA



SQL / DB



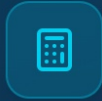
Power BI



Tableau



Web Scraping



STAGE 01 • SCIENTIFIC FOUNDATION

Math & Statistics

Milestone 01 / 08

OBJECTIVE & CORE FOCUS

Master statistical distributions, probability theory, hypothesis testing, and variance analysis to uncover the mathematical truth behind data.



HANDS-ON DELIVERABLE:

Statistical evaluation report assessing data health and business hypotheses.

KEY COMPETENCIES & TOOLS

• Descriptive Statistics

• Probability & Distributions

• Hypothesis Testing (A/B)

• Outlier Detection

SECTION 01

Excel for Data Analysis

The foundation every data analyst starts with.



PART 001

— Core Strengths of Excel in Analytics

Why Excel is Important

It's the most accessible tool to start organizing and exploring data.



Data Entry & Organization

Structure raw information into clean rows and columns.



Formulas & Functions

Automate calculations across large datasets with precision.



Data Cleaning

Remove duplicates, fix errors, and standardize formats.



Pivot Tables

Summarize and cross-tabulate large datasets instantly.



Charts

Turn tables into visual, easy-to-read data stories.



Dashboards

Combine multiple visuals into one interactive view.

Everyday Analytical Functions

Formulas & Functions in Practice

A handful of functions cover most everyday analysis needs.

SUM

Calculate Total Sum

fx

AVERAGE

Compute Arithmetic Mean

fx

IF

Conditional Logic & Tiering

fx

XLOOKUP

Modern Search & Relation...

fx

Sheet1

fx

=XLOOKUP("ORD-104", A2:A6, B2:B6)

#	A (Order ID)	B (Customer)	C (Sales Revenue)	D (Loyalty Tier)
2	ORD-101	Sarah Connor	\$1,200	Regular
3	ORD-102	Alex Mercer	\$2,450	VIP
4	ORD-103	Elena Rostova	\$850	Regular
5	ORD-104	Michael Scott	\$2,150	VIP
6	ORD-105	David Kim	\$1,200	Regular

LIVE DYNAMIC OUTPUT

Modern Search & Relational Lookup

Modern flexible replacement for VLOOKUP without left/right lookup limits.

Lookup Return Value

Michael Scott

Pivot Tables aggregate and group these calculations across millions of rows with drag & drop!

End-to-End Excel Pipeline

From Clean Data to Dashboards

The final step: turning a clean table into a visual story.

01

Data Cleaning & Prep

Active Stage

02

Build Pivot Tables

Click to inspect

03

Insert Pivot Charts

Click to inspect

04

Connect Slicers & Timeline

Click to inspect

STAGE 01 / 04 STEP 1

Data Cleaning & Prep

Remove duplicates, handle blanks, and format ranges into official Excel Tables (Ctrl+T).

- ✓ Convert data range into Table (Ctrl+T)
- ✓ Remove duplicate entries automatically
- ✓ Standardize dates, currencies, and numbers

Milestone Target:

100% Cleaned Data



Excel Workspace Simulation

Live Flow

Raw Records

15,420 rows

Nulls Cleaned

824 fixed

Ready Dataset

14,596 rows

Previous Step

1 of 4

Next Step

SECTION 02

Python Fundamentals

The programming language that powers modern data analysis.



PART 002

Core Building Blocks in Code

Variables & Data Types

Python's role: automating analysis beyond what Excel can handle.



Variables & Fundamental Data Types

The core building blocks for storing and processing data in Python

- int / float: Store integer and decimal numbers (prices, quantities, metrics)
- str: Store text strings (customer names, categories, and geographic locations)
- bool: Store logical True/False flags for boolean conditions and filters
- type(): Built-in Python function to inspect and validate variable data types

Interactive Code Implementation

PYTHON



PYTHON SCRIPT

Copy Code

```
# Customer Transaction Record
customer_name = "Sarah Johnson"      # str
order_id = 10482                      # int
total_amount = 249.75                # float
is_loyalty_member = True              # bool

# Compute loyalty discount dynamically
discount = 0.15 if is_loyalty_member else 0.0
final_price = total_amount * (1 - discount)

print(f"Customer: {customer_name}")
print(f"Order #{order_id} Total: ${final_price:.2f}")
```

>_ Ready to execute in Jupyter / VSCode / SQL Console

• Valid

— Branching & Iteration

Conditions & Loops

Logic that lets code react to data and repeat tasks automatically.

PYTHON LOGIC & ITERATION ENGINE:

[Core Pillars \(PDF View\)](#)[🌟 Interactive Lab](#)

BRANCHING LOGIC

Conditions

if / elif / else

- if / elif / else evaluate a condition
- Used to filter, flag, or branch logic
- Example: flag rows where revenue < target

```
if row['revenue'] < target:  
    row['flag'] = 'Under Target'
```



ITERATIVE LOGIC

Loops

for / while

- for & while loops repeat code block
- Iterate over datasets row by row
- Example: process all customer records

```
for client in dataset:  
    process_record(client)
```

Conditions and loops form the fundamental pillars of data pipeline automation in Python

Reusable Data Structures & Logic

Functions, Lists & Dictionaries

Structures that organize data and logic for reuse.



Functions, Lists & Dictionaries

Data structures and reusable modular logic for automation

- Lists []: Ordered arrays to store metric sequences (e.g. daily sales records)
- Dictionaries {}: Key-value pairs representing structured entity profiles
- Functions def: Modular functions to automate repetitive analytical formulas
- Loops for / while: Iterate through datasets to clean and transform records

 Interactive Code Implementation

PYTHON



PYTHON SCRIPT

 Copy Code

```
# 1. List of Daily Sales Totals
daily_sales = [1200, 1850, 2100, 950, 3200]

# 2. Dictionary of Customer Entity
customer = {
    "id": "C-901",
    "name": "Ahmed Hassan",
    "city": "Cairo",
    "orders_count": 12
}

# 3. Reusable KPI Calculation Function
def calculate_growth(current, previous):
    rate = ((current - previous) / previous) * 100
    return round(rate, 2)
```

SECTION 03

Python for Data Analysis

Putting Python libraries to work on real datasets.



PART 003

The Two Indispensable Libraries

NumPy & Pandas

The two libraries every data analyst uses daily.

PYTHON CORE DATA STACK ECOSYSTEM:

Dual Comparison

NumPy Engine

Pandas DataFrames



NUMERICAL FOUNDATION

NUMPY

```
import numpy as np
```

- Fast numerical operations
- Works with arrays and matrices
- Foundation for other data libraries
- Powers statistical calculations

 N-Dimensional ndarray

[Explore Engine →](#)



TABULAR MANIPULATION

PANDAS

```
import pandas as pd
```

- Data manipulation and analysis
- DataFrames organize rows and columns
- Filter, group, and merge datasets easily
- Reads CSV, Excel, SQL and more

 2D Tabular DataFrames

[Explore DataFrames →](#)

NumPy delivers high-speed numerical array crunching, while Pandas provides flexible tabular DataFrames

Python Visualizations

Matplotlib & Seaborn

Python's visualization toolkit for charts and statistical plots.

PYTHON'S VISUALIZATION TOOLKIT:

Dual Comparison

Matplotlib Engine

Seaborn Statistical Lab



BASE PLOTTING ENGINE

Matplotlib

```
import matplotlib.pyplot as plt
```

The base plotting library — full control over every chart element: line plots, bar charts, histograms.

- Fine-grained control over axes, ticks, colors, labels
- Standard charts: Line, Bar, Histograms, Scatter
- Object-oriented Figure and Subplots architecture

PLOTS:

Line

Bar

Hist

[Interactive Demo →](#)


STATISTICAL VISUALS

Seaborn

```
import seaborn as sns
```

Built on Matplotlib for statistical visualization — cleaner defaults for distributions, correlations, and comparisons.

- Correlation heatmaps and multi-variable matrices
- Kernel Density Estimation (KDE) and distribution curves
- Native DataFrame integration with high-level API

STATS:

KDE

Heatmap

Box

[Interactive Demo →](#)

Matplotlib gives you low-level architectural control, while Seaborn delivers effortless statistical beauty



Real-World Data Workflow with Pandas

From importing raw files to extracting key statistical summaries

- `pd.read_csv()`: Ingest massive datasets directly into high-speed DataFrames
- `df.dropna()`: Handle missing values and filter incomplete records
- `df.groupby()`: Segment performance metrics by category and branch location
- `df.describe()`: Compute complete statistical profiles in a single command



Interactive Code Implementation

PYTHON

SECTION 04

SQL & Databases

The language of structured data — used in nearly every company.



PART 004

Relational Data Storage

Why Databases Matter

Where structured business data actually lives.

RELATIONAL DATABASE ARCHITECTURE:

[Core Pillars \(PDF View\)](#)
[Interactive Schema](#)


ACID Safe

STORAGE ENGINE

Database Concepts

A structured system for storing and retrieving data reliably.

✓ Integrity & Concurrency



Rows & Cols

TABULAR SCHEMA

Tables

Data organized into rows (records) and columns (fields).

📄 Standardized Schema



PK → FK

RELATIONAL KEYS

Relationships

Tables connect through shared keys, e.g. customer ID.

🔑 Primary & Foreign Keys

[View Link](#)

Databases are where enterprise business data lives — relational keys are what give them power

— Querying & Slicing Tables

SQL Queries & Filtering

Ask precise questions of your data using SQL.



Core SQL Querying Architecture

Extracting, filtering, and ordering tabular records from RDBMS

- **SELECT / FROM:** Specify columns to retrieve and the primary source table
- **WHERE:** Filter records using conditional criteria (amount > 1000, active status)
- **GROUP BY / HAVING:** Aggregate rows and apply aggregate thresholds (SUM, COUNT)
- **ORDER BY / LIMIT:** Sort records ascending/descending and isolate top performers

Interactive Code Implementation

SQL



SQL SCRIPT

Copy Code

```
-- Top 5 High-Value Customers in 2024
SELECT
    c.customer_id,
    c.customer_name,
    c.city,
    COUNT(o.order_id) AS total_orders,
    SUM(o.order_amount) AS total_spent
FROM customers c
JOIN orders o ON c.customer_id = o.customer_id
WHERE o.order_date >= '2024-01-01'
GROUP BY c.customer_id, c.customer_name, c.city
HAVING SUM(o.order_amount) > 5000
ORDER BY total_spent DESC
LIMIT 5;
```

> Ready to execute in Jupyter / VSCode / SQL Console

• Valid

Summaries & Multi-Table Joins

Aggregation & Joins

Summarize data and combine multiple tables together.



Relational Joins (INNER, LEFT, FULL)

Combining related records across tables using Primary & Foreign Keys

- **INNER JOIN:** Returns only rows with matching foreign keys in both tables
- **LEFT JOIN:** Keeps all rows from left table, with NULLs for unmatched right rows
- **RIGHT JOIN:** Keeps all rows from right table with corresponding left matches
- **FULL OUTER JOIN:** Combines all records from both tables regardless of match

Interactive Code Implementation

SQL



SQL SCRIPT

Copy Code

```
-- Left Join to Find Inactive / Non-Purchasing Customers
SELECT
    c.customer_id,
    c.customer_name,
    c.signup_date,
    o.order_id,
    o.order_amount
FROM customers c
LEFT JOIN orders o ON c.customer_id = o.customer_id
WHERE o.order_id IS NULL; -- Identifies registered users with 0 purchases
```

> Ready to execute in Jupyter / VSCode / SQL Console

Valid

Nested Queries & Database Engines

Subqueries & SQL Tools

Nesting queries for advanced logic, and where SQL runs.

Subqueries, CTEs & Window Functions



Nested queries, Common Table Expressions, and analytical rankings

- Subqueries: Nested SELECT statements to evaluate dynamic scalar thresholds
- CTEs (WITH clause): Reusable named query blocks for clean readability
- DENSE_RANK() OVER(): Compute rank hierarchies per category partition
- PARTITION BY: Execute windowed aggregations without collapsing row granularity

Interactive Code Implementation

SQL

SQL SCRIPT

Copy Code

```

-- Rank Top Selling Products within Each Category
WITH RankedProducts AS (
    SELECT
        category_name,
        product_name,
        revenue,
        DENSE_RANK() OVER (
            PARTITION BY category_name
            ORDER BY revenue DESC
        ) AS rank_in_category
    FROM sales_records
)
SELECT * FROM RankedProducts
WHERE rank_in_category <= 3;
    
```

>_ Ready to execute in Jupyter / VSCode / SQL Console

Valid

SECTION 05

Power BI

Turning models into business-ready dashboards.



PART 005

— ETL & Star Schema Modeling

Power Query & Data Modeling

Power BI role in business intelligence: connect, shape, and model data.

POWER BI DATA PREPARATION & ARCHITECTURE:

Core Pillars (PDF View)

Power Query ETL

Star Schema Model



ETL Engine

DATA INGESTION

Power Query

Connect to sources and clean/shape data before loading it in.

Applied Steps

View ETL



Star Schema

LOGICAL DESIGN

Data Modeling

Organize tables into a logical model for analysis.

Facts & Dimensions

View Schema



1 : Many (1:*)

MODEL CARDINALITY

Relationships

Connect tables together using shared keys.

✓ Cross-Filtering Active

Power Query cleans and shapes data; Data Modeling builds the engine for lightning-fast business insights

Power BI Calculation Language

DAX — Data Analysis Expressions

The formula language behind Power BI calculations.

DAX Measures & Context Transition in Power BI

Authoring dynamic business KPIs and Time Intelligence calculations

- **CALCULATE():** Core DAX engine function that overrides and manipulates filter context
- **Time Intelligence:** Compare period performance (SAMEPERIODLASTYEAR, YTD)
- **Measures:** Computed dynamically on-the-fly to optimize RAM and speed
- **DIVIDE():** Safe mathematical division preventing division-by-zero crashes

Interactive Code Implementation

DAX

DAX SCRIPT

Copy Code

```
-- 1. Dynamic Year-over-Year Revenue Growth
Total Revenue = SUM(Sales[Revenue])

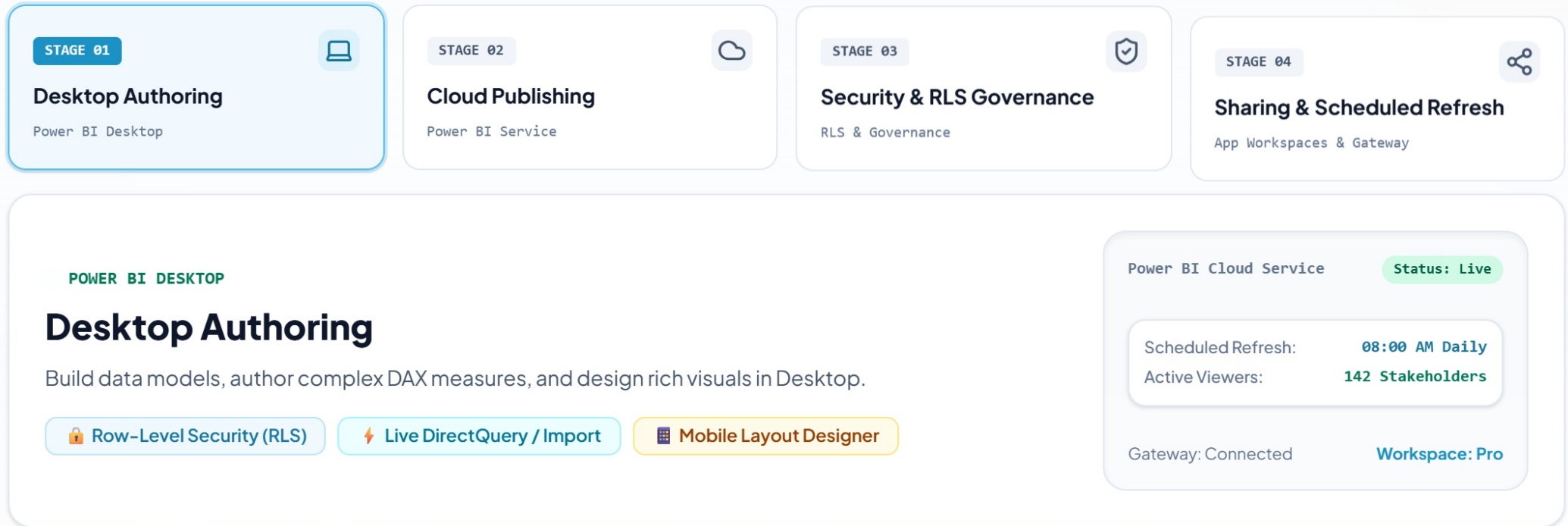
Revenue LY =
CALCULATE(
    [Total Revenue],
    SAMEPERIODLASTYEAR('Calendar'[Date])
)

YoY Growth % =
DIVIDE(
    [Total Revenue] - [Revenue LY],
    [Revenue LY],
    0
)
```

— From Model to Shared Workspace

Dashboards, Reports & Publishing

From a data model to a shareable business tool.



SECTION 06

Tableau

A leading tool for visualization and data storytelling.



Visual Analytics Platform

Connecting Data & Creating Charts

Tableau as a visualization and business intelligence tool.



COMPONENT 01

Connecting Data Sources

Plug directly into spreadsheets, relational databases, and cloud data warehouses.

Visual Analytics

01



COMPONENT 02

Creating Charts

Drag-and-drop chart building with instant interactive visual feedback.

Visual Analytics

02

Engaging Stakeholder Communication

Interactive Dashboards & Storytelling

Tableau's real strength: making data explorable and memorable.



COMPONENT 01

Interactive Dashboards

Filters and actions let viewers explore the data dimensions themselves.

Visual Analytics

01



COMPONENT 02

Storytelling with Data

Sequence dashboards into a guided narrative for executive stakeholders.

Visual Analytics

02

— BI Industry Giants Comparison

Power BI vs Tableau

Two leading BI tools — often complementary, not competing.



PILLAR 01

POWER BI

Tight Microsoft ecosystem integration and powerful DAX modeling.

- Tight Microsoft ecosystem integration
- Strong data modeling with DAX
- Cost-effective licensing
- Great for business reporting

Core Concept

01



PILLAR 02

TABLEAU

Best-in-class visual flexibility and exploratory drag-and-drop analysis.

- Best-in-class visual flexibility
- Deep, intuitive drag-and-drop analysis
- Popular for storytelling & exploration
- Strong for advanced visual analytics

Core Concept

02

SECTION 07

Web Scraping

Collecting data directly from the web when no dataset exists.



PART 007

— Automated HTML Extraction

What is Web Scraping?

Extracting data automatically from websites and web pages.



HTML Structure

Web pages are built from tags — scraping parses that structural hierarchy.

● System Standard



Requests

A Python library used to fetch a web page's raw HTML content over HTTP.

● System Standard



BeautifulSoup

Parses HTML DOM trees so specific elements and text can be extracted.

● System Standard



APIs

A cleaner, structured alternative to scraping raw HTML pages.

● System Standard

Real-world Scraping Applications

Web Scraping in Practice

Real scenarios where analysts use scraping to fill data gaps.

WEB SCRAPING REAL-WORLD SCENARIOS:

[Core Pillars \(PDF View\)](#)
[Price Tracker Lab](#)
[Scheduled Automation](#)


E-Commerce

COMPETITIVE INTEL

Collecting Product Prices

Track competitor pricing across e-commerce sites.

Dynamic Pricing

[Try Scraper](#)


Public Data

MARKET RESEARCH

Market Analysis

Aggregate public data to study market trends.

[Trend Aggregation](#)


Cron Job

SCHEDULED SCRIPTS

Data Collection Automation

Schedule scripts to gather fresh data on a routine.

Cron Pipelines

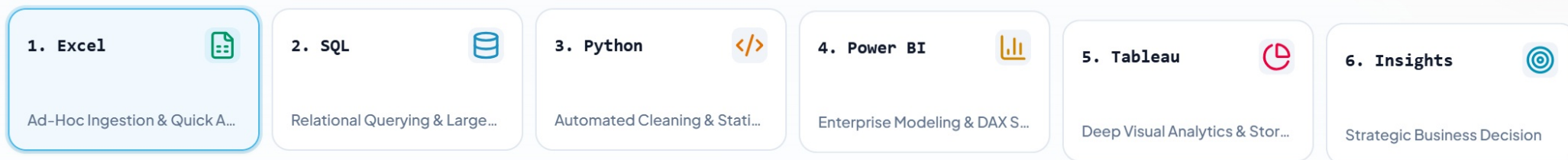
[View Schedule](#)

Web scraping fills critical data voids when no formal dataset or public API exists

— End-to-End Tool Pipeline

Complete Data Analyst Workflow

How every tool in this roadmap fits into one pipeline.



STAGE 01 / 06

AD-HOC INGESTION & QUICK AUDITS

Clean Tabular Spreadsheets

1. Excel: Ad-Hoc Ingestion & Quick Audits

Ingest raw sheets, perform ad-hoc checks, and run rapid initial summaries via Pivot Tables.

✨ Harmonious Synergy: Each tool amplifies the next in a modern enterprise stack!

Previous Tool

Next Tool

— The Full Skill Stack

Career Skills of a Data Analyst

Technical ability and soft skills both matter for the job.

 THE JOB-READY EQUATION (50% TECHNICAL + 50% BUSINESS & SOFT SKILLS):

Complete Matrix

Technical (50%)

Soft Skills (50%)



HARD SKILLS • 50% WEIGHT

Technical Skillset

Code & Tools

Excel & Advanced Spreadsheets

Formulas, Pivot Tables, Data Cleansing

Core

SQL & Database Architecture

Queries, Joins, Aggregation, Subqueries

Top In-Demand

Python for Data Analysis

NumPy, Pandas, Matplotlib, Seaborn

Automation

BI Dashboards (Power BI / Tableau)

DAX Modeling, Power Query, Storytelling

Business Facing



SOFT & BUSINESS • 50% WEIGHT

Business & Soft Skills

Strategy & Value

Business Acumen

Connecting metrics directly to revenue and business goals

Strategy

Data Storytelling & Communication

Translating complex findings into clear executive actions

Influence

Problem Solving & Logic

Breaking down ambiguous business questions logically

Thinking

Critical Thinking

Questioning assumptions and validating data veracity

Quality

— Complete Session Recap

Session Summary

Here's everything we covered in this session.



What is Data Analysis?

Collecting, cleaning, analyzing, and interpreting data.



Data Types

Qualitative, quantitative, structured, and unstructured.



Data Pipeline

Sources → Extract → Transform → Load → Analysis.



Analysis Lifecycle

Collect, clean, explore, analyze, visualize, communicate.



Data Analyst Roadmap

Excel → Python → SQL → Power BI → Tableau → Web Scraping.



Required Tools

Excel, Python, SQL, Power BI, Tableau, and scraping basics.

— Interactive Self-Assessment

Knowledge Check

Test your understanding before moving to the next session.

1 1. What is the difference between structured & unstructured data?

Click to reveal 💡

2 2. What are the 6 steps of the Data Analysis Lifecycle?

Click to reveal 💡

3 3. Why do we need SQL when we already have Excel?

Click to reveal 💡

4 4. What is the main strength of Power BI vs Tableau?

Click to reveal 💡

5 5. Which Python library is used for manipulating DataFrames?

Click to reveal 💡

6 6. What is the difference between Data, Information, and Insight?

Click to reveal 💡

— Practical Research Assignment

Assignment — Mini Research Task

Choose a company and analyze how it uses data.

 SELECT A COMPANY FOR YOUR CASE STUDY ASSIGNMENT:

Netflix

Amazon

Uber

Spotify

SUBMISSION REQUIREMENTS

1-Page Summary Report

The 4 Questions to Answer for (Netflix):

- 1 What specific data types does this company collect daily?
- 2 How does the company transform raw data into actionable insights?
- 3 What critical business decisions are driven by their analytics models?
- 4 Which tools from our diploma roadmap (SQL, Python, Power BI) fit best?

Due Date: Before Session 02 for peer discussion

Practical Task



Netflix

Streaming & Entertainment



DATA COLLECTED:

Watch history timestamps, pause/resume points, search queries, completion rates, and device telemetry.



DRIVEN BUSINESS DECISIONS:

Personalized thumbnail generation, content recommendation engine, and budgeting \$17B original shows.

Reference Case

[Netflix Case](#)

[i]NSTANT

✓ SESSION 01 COMPLETED SUCCESSFULLY

THANK YOU

Thank you for your active participation! Get ready for hands-on Excel data mastery in the next session.

[↺ Restart Presentation](#)[Open Session 02: Excel Basics →](#)